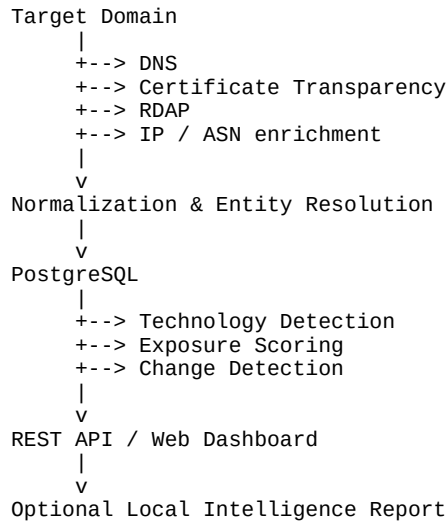


SurfaceIntel

Passive-first OSINT-driven external attack surface intelligence in Go

1. Project vision

SurfaceIntel is a Go-based OSINT and external attack-surface intelligence platform. The goal is to discover, normalize, enrich and track publicly observable infrastructure rather than simply run another generic port scanner.



2. MVP

Keep version 0.1 deliberately small. The first useful command should be:

```
surfaceintel scan example.com
```

Discover DNS records, Certificate Transparency names, IP addresses and ASN information, normalize them and store the results in PostgreSQL with first-seen and last-seen timestamps.

3. Data sources

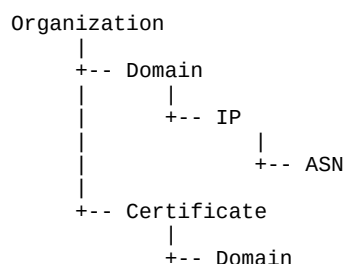
DNS: A, AAAA, CNAME, MX, NS and TXT records. Start with Go's `net.Resolver`.

Certificate Transparency: use public CT data such as `crt.sh` to discover additional hostnames.

RDAP / ASN: enrich IP addresses with registration and network information.

4. Entity resolution

The key design challenge is recognizing that domains, IPs, certificates and ASNs are related observations of the same infrastructure.



5. Go architecture

```

surfaceintel/
+-- cmd/surfaceintel/main.go
+-- internal/
|   +-- dns/
|   +-- crtsh/
|   +-- rdap/
|   +-- asn/
|   +-- enrichment/
|   +-- discovery/
|   +-- database/
|   +-- scoring/
|   +-- api/
+-- web/
+-- migrations/
+-- docs/
+-- Dockerfile
+-- docker-compose.yml
+-- go.mod
+-- README.md

```

Go's concurrency model is a natural fit because DNS, CT, RDAP and enrichment sources have different latency characteristics.

6. Database

Start with PostgreSQL rather than Neo4j. Suggested tables: organizations, domains, ip_addresses, domain_ips, certificates, certificate_domains, asns, technologies, assets and observations.

The observation model is especially important: store what was observed, when it was first seen, when it was last seen and where it came from. This enables historical intelligence.

7. Technology detection

Add lightweight HTTP and TLS enrichment after passive discovery. Detect technologies using headers, common paths and HTML/JavaScript indicators. Keep fingerprints explainable and source-backed.

8. Exposure Score

Use an explicitly documented Exposure Score, not a vulnerability score. Possible inputs include exposure characteristics, technology, environment, age, certificate state, infrastructure context and intelligence indicators. The score should explain why an asset was ranked.

9. Historical intelligence

Track new subdomains, new IPs, disappearing assets, certificates and technology changes. This turns a one-time reconnaissance tool into a continuous external attack-surface intelligence system.

```

RECENT CHANGES

+ staging2.example.com
+ 203.0.113.51
- old-api.example.com
+ new TLS certificate
~ technology changed

```

10. Web interface

A Go + HTMX dashboard is a strong initial choice. Useful pages: Overview, Assets, Asset Detail, Certificates, Infrastructure, Technologies, Recent Changes and Intelligence Reports.

11. Local AI

Once the structured pipeline works, integrate a local Qwen model through Ollama. Give it structured observations and ask it to produce an intelligence report. Explicitly separate OBSERVED, INFERRED and UNKNOWN so that the LLM is

not treated as the source of truth.

12. Roadmap

- v0.1 Go CLI, DNS, CT, IP resolution, PostgreSQL
- v0.2 RDAP, ASN enrichment, concurrency, HTTP/TLS metadata
- v0.3 Technology detection, Exposure Score, history, change detection
- v0.4 REST API, Go + HTMX dashboard, Docker Compose, CI/tests
- v0.5 Threat-intelligence enrichment, local Qwen reporting, case study

13. Responsible use

Design the project for assets you own or are authorized to assess. Keep the initial discovery passive-first and avoid aggressive scanning of third-party infrastructure. For public demonstrations, use your own infrastructure, a lab, deliberately provided targets or appropriate test domains.

14. LinkedIn / portfolio positioning

Focus the public case study on the problem and engineering decisions: how DNS, CT, IP/ASN and HTTP observations are normalized into an asset model, how changes are detected, and how evidence-backed intelligence reports are generated.

15. Name

SurfaceIntel is the recommended name. It is short, professional and broad enough to cover OSINT, attack-surface management, threat intelligence and later local-AI reporting.